

1 |

internet, une infrastructure sociotechnique

thomas tari

- 1 1 | une révolution numérique
- 1 2 | philosophie des réseaux
- 1 3 | communautés pionnières
- 1 4 | penser l'infrastructure
- 1 5 | enjeux sociotechniques

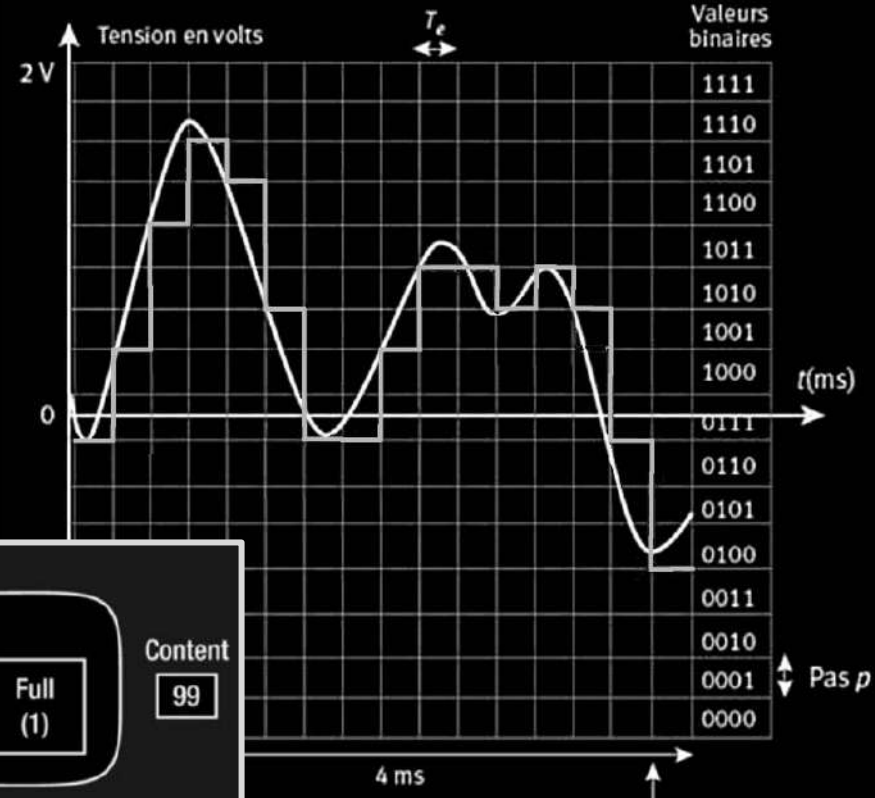
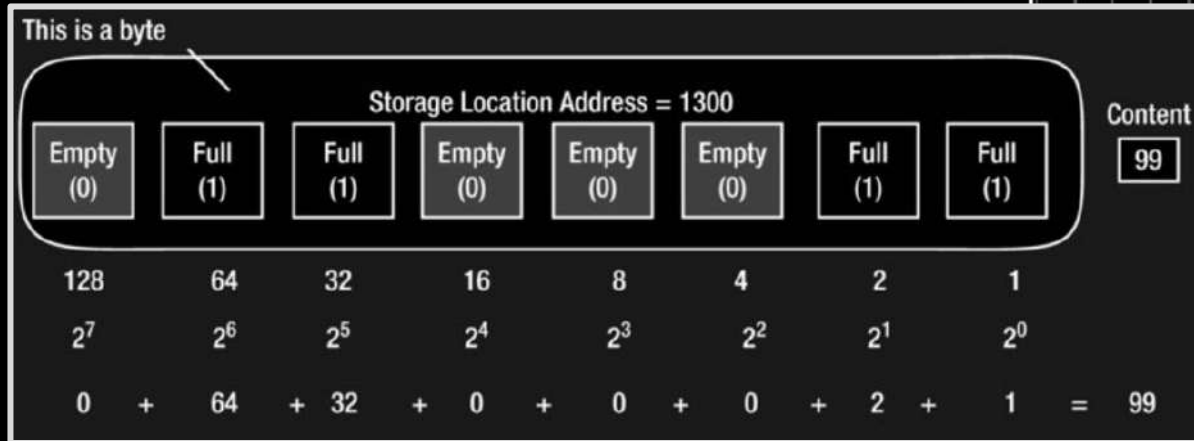
10 sep 2026

1 1 | une révolution numérique

signal analogique | | signal numérique

- intégrité des données
- vitesse de transmission

→ **accélération** du transfert
d'information et meilleure **qualité**



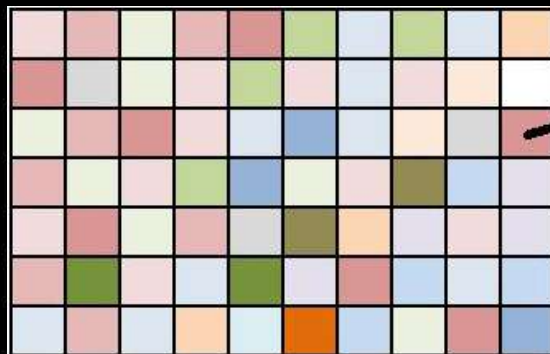
→ quelles
conséquences ?

Decimal	Hex	Char	Decimal	Hex	Char
64	40	@	96	60	`
65	41	A	97	61	a
66	42	B	98	62	b
67	43	C	99	63	c
68	44	D	100	64	d
69	45	E	101	65	e
70	46	F	102	66	f
71	47	G	103	67	g
72	48	H	104	68	h
73	49	I	105	69	i
74	4A	J	106	6A	j
75	4B	K	107	6B	k
76	4C	L	108	6C	l
77	4D	M	109	6D	m
78	4E	N	110	6E	n
79	4F	O	111	6F	o
80	50	P	112	70	p
81	51	Q	113	71	q
82	52	R	114	72	r
83	53	S	115	73	s
84	54	T	116	74	t
85	55	U	117	75	u
86	56	V	118	76	v
87	57	W	119	77	w
88	58	X	120	78	x
89	59	Y	121	79	y
90	5A	Z	122	7A	z
91	5B	[	123	7B	{
92	5C	\	124	7C	
93	5D	]	125	7D	}
94	5E	^	126	7E	~
95	5F	_	127	7F	[DEL]

← table
ascii

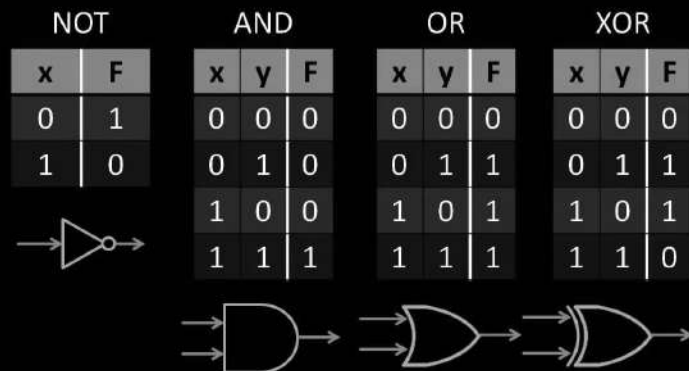
opérateurs
booléens →

couleurs rgb

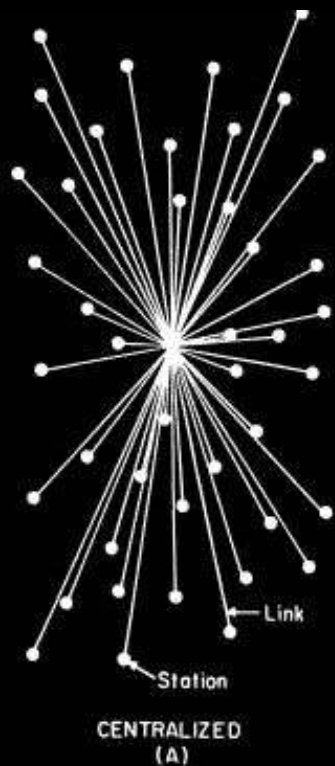


RGB (218, 150, 149)

R = 11011010
G = 10010110
B = 10010101

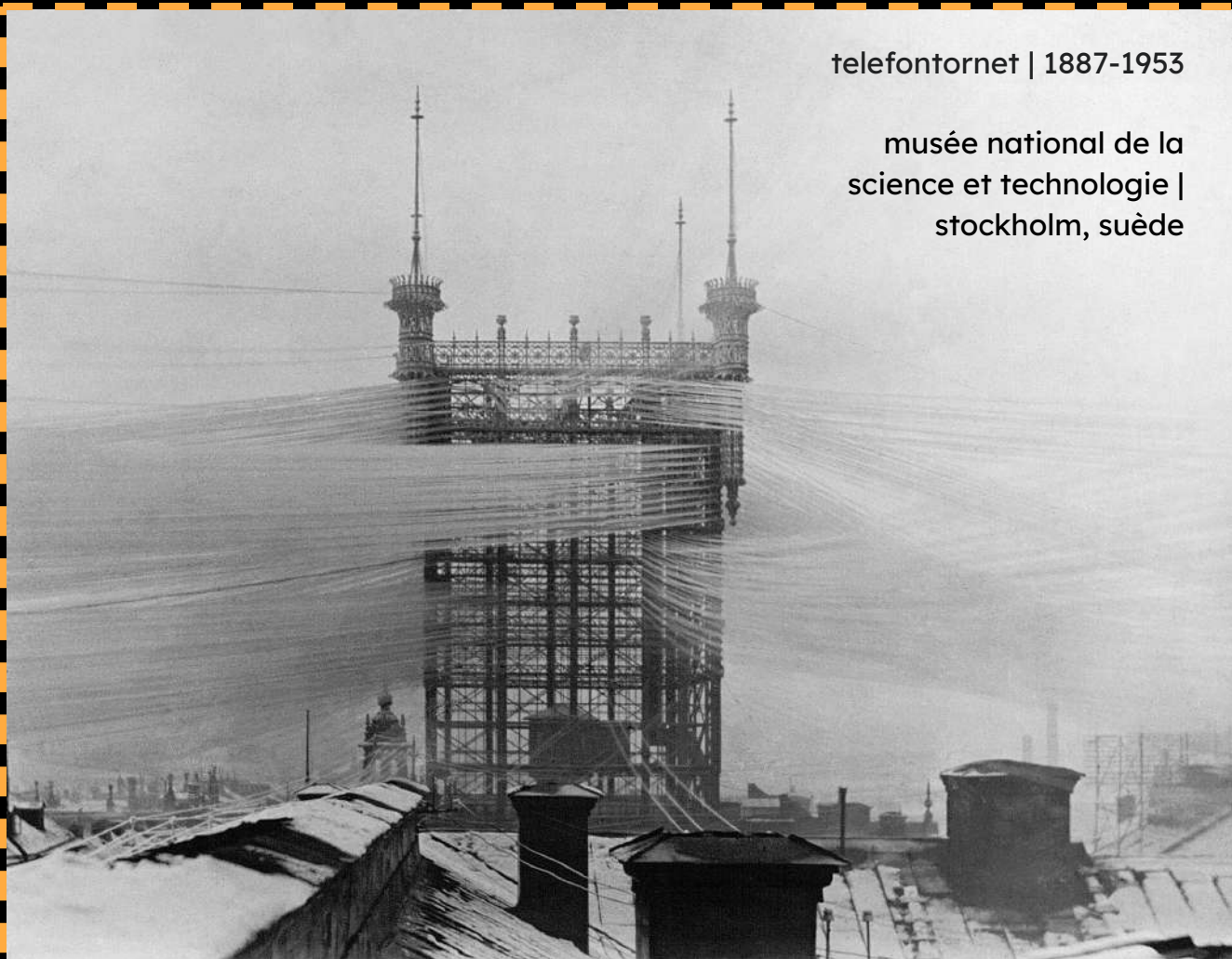


1 2 | philosophie des réseaux



telefontornet | 1887-1953

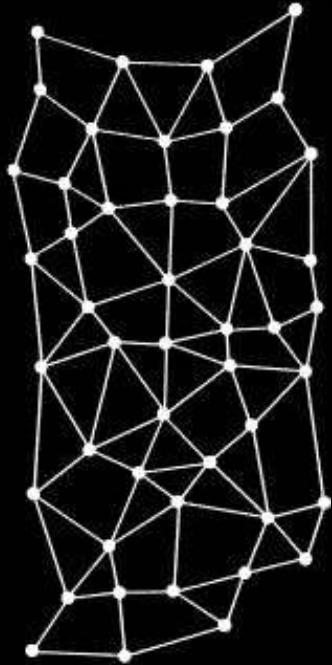
musée national de la
science et technologie |
stockholm, suède



arpa (advanced research project agency) - net (1969)

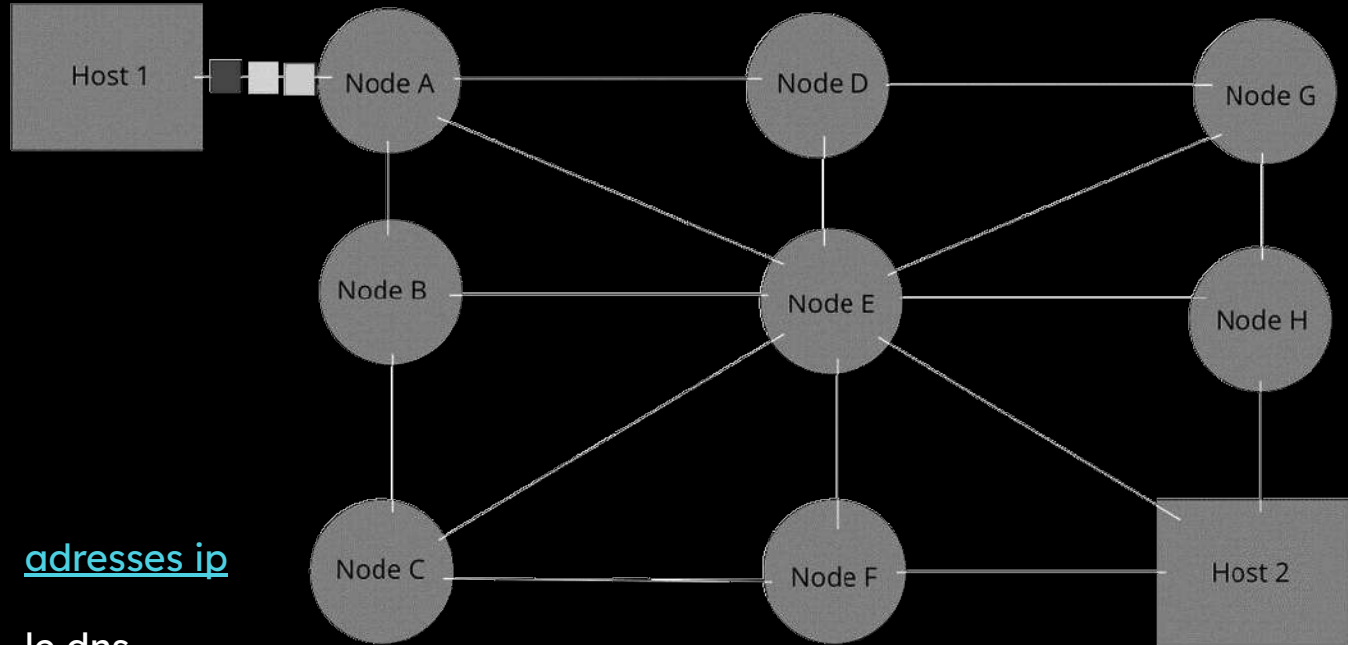


tcp/ip, paquets réseaux, requests for comments (1969-83)



DISTRIBUTED
(C)

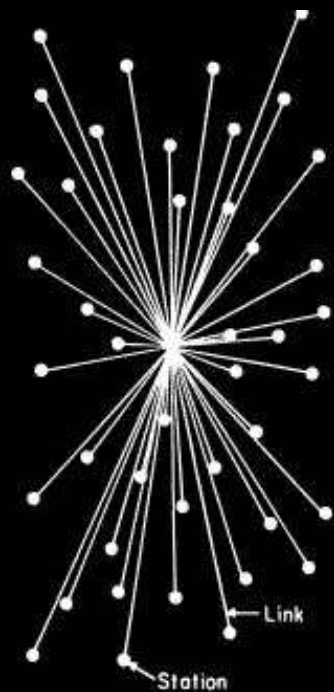
The original message is Green, Blue, Red.



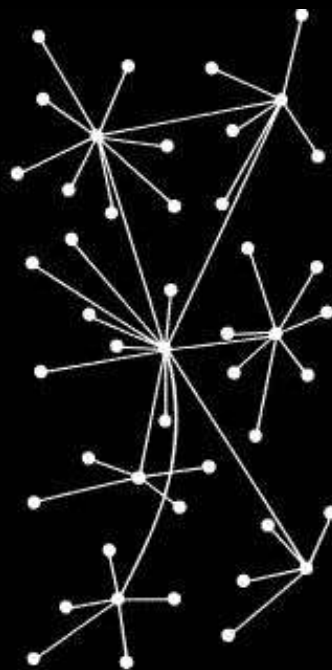
adresses ip

le dns

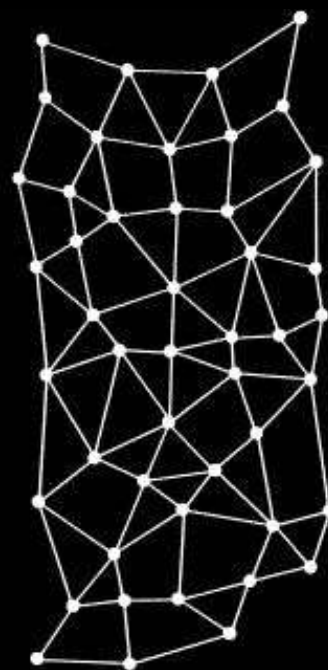
un réseau de réseaux



CENTRALIZED
(A)

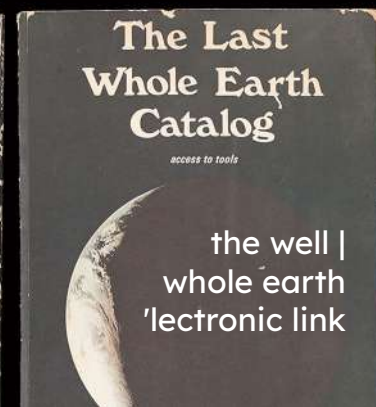
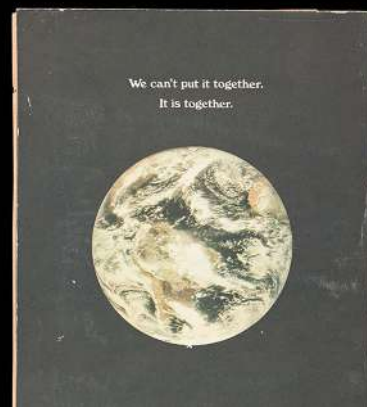
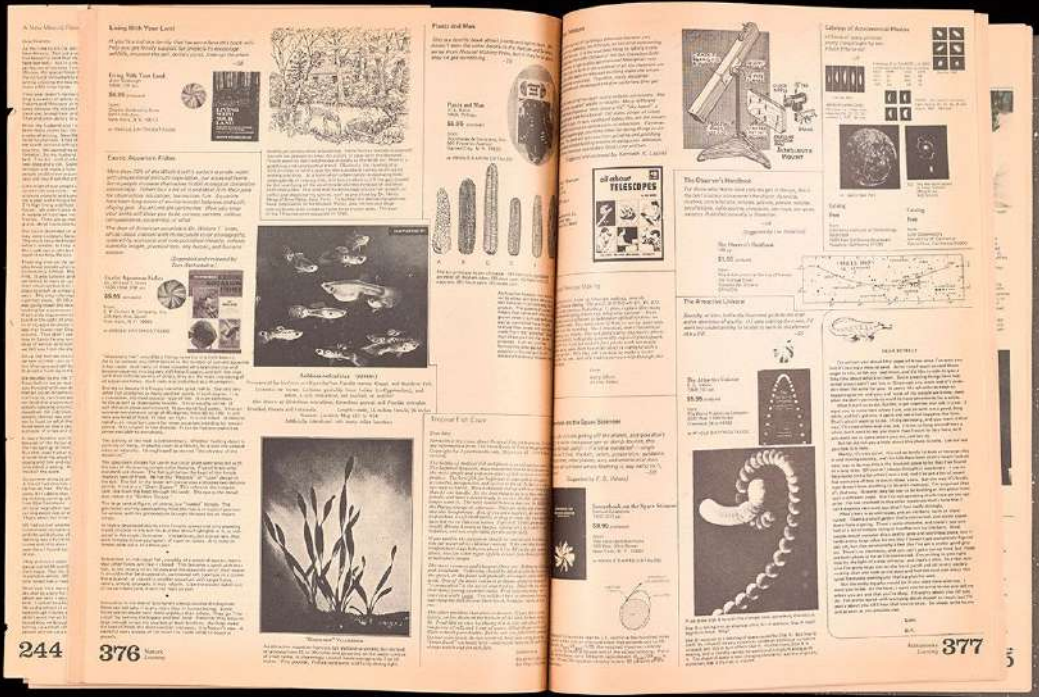


DECENTRALIZED
(B)



DISTRIBUTED
(C)

1 3 | communautés pionnières



le premier internet : liberté,
créativité et homogénéité

portraits d'informaticiennes



ada lovelace
(1815-1852)

grace hopper
(1906-1992)



frances allen
(1932-2020)



margaret hamilton
(1936-)

les 'human computers'



katherine johnson
(1918-2020)



mary jackson
(1921-2005)

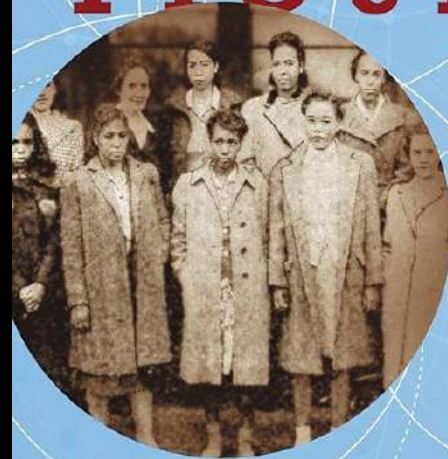


dorothy vaughan
(1910-2008)

THE #1 NEW YORK TIMES BESTSELLER

During World War II, America's fledgling aeronautics industry hired black female mathematicians to fill a labor shortage. These "human computers" stayed on to work for NASA and made sure America won the Space Race. They fought for their country's future, and for their share of the American dream. This is their untold story.

HIDDEN FIGURES



MARGOT LEE SHETTERLY

comment elles ont été exclues



déquali-
fication
(1960s)

pénurie de
main d'
œuvre
(mi-1960s)

This is all the staff you need to process orders, produce invoices, check credit, check and analyse sales, check stocks, produce despatch notes and operate a computer.

INTERACT 75 makes it possible.
The Baric INTERACT 75 service gives you direct access to a major computer system from your own office. Easily, Quickly. And economically.

But the real advantage of INTERACT 75 is the fact that the computer you plug into is already programmed to carry out the day to day commercial processes involved in running your business. Like recording invoices. Issuing orders. Updating stock records. Checking customer credit limits. And calculating discounts.

In the opening phase of INTERACT 75 we are offering this Commercial Management Service to a limited number of customers.

These will probably be companies with about 200-2,000 personnel; selling from stock; and coping with an invoicing problem.

Briefly, this is how it works. All your orders, from whatever source, are passed to the girl operator at your INTERACT 75 terminal. She transfers the order data to the computer, via the terminal. The computer immediately checks

all the data against permanent file information. If something is amiss, such as an item being out of stock, it advises her immediately. If not, her single entry starts a number of balls rolling. The computer records all the information on the user's private file: quantity, ledger entry, customer's name and address and so on, allowing for later detailed analysis for management reports, stock control, etc.

The computer then scans the file to prepare the invoice, noting any discount arrangements and adjusting the account. Within seconds it passes the information back to the user's office where it prints out the completed invoice automatically. The implications are obvious to every businessman.

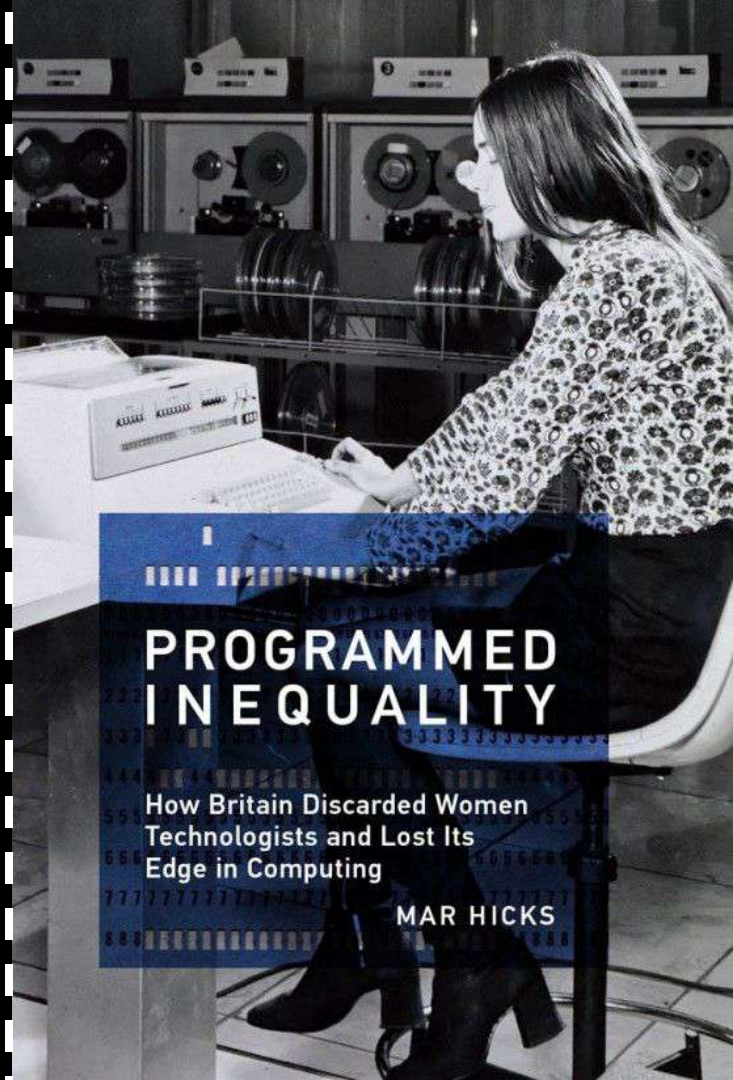
Unfortunately we can only serve a few. So we would urge you to be quick about writing for details to: Baric Computing Services Limited, 68 Newman Street, London W1P 4EH. An ICL/Barclays Company.

BARIC

INTERACT 75 Europe's first commercial time sharing system.



transformation genrée du
travail et exportation coloniale
des normes (1970s)



PROGRAMMED INEQUALITY

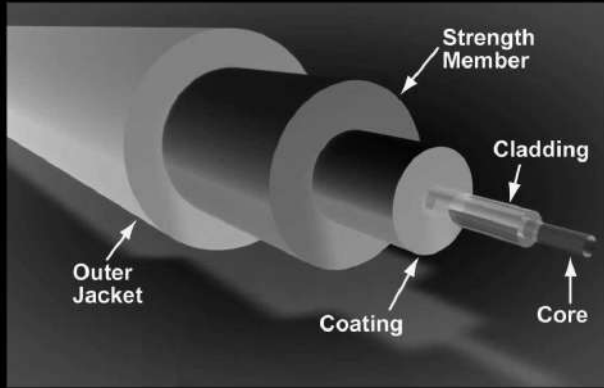
How Britain Discarded Women
Technologists and Lost Its
Edge in Computing

MAR HICKS

1 4 | penser l'infrastructure

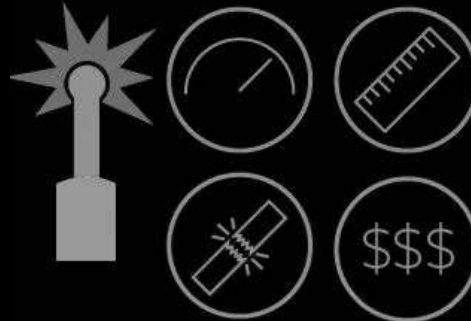
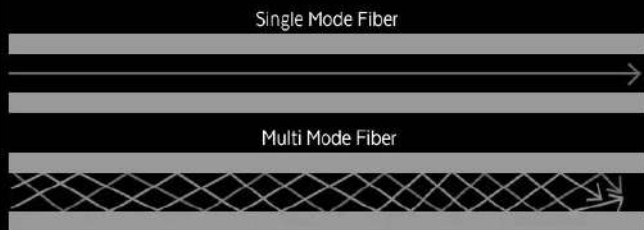
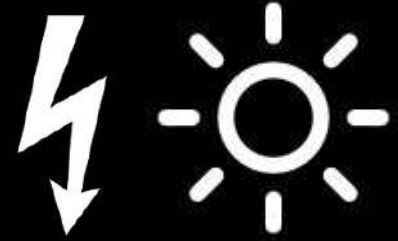
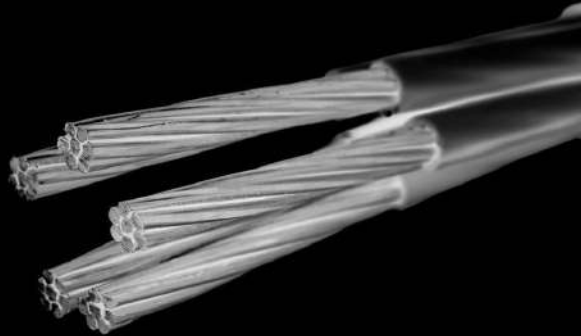
la matérialité de l'infrastructure : des câbles

Fibre optic Cable

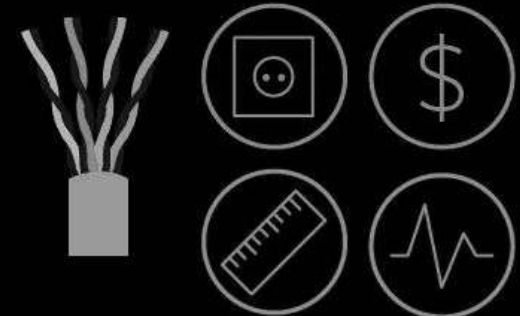


V/s

Copper Cable



COPPER



la matérialité des hubs

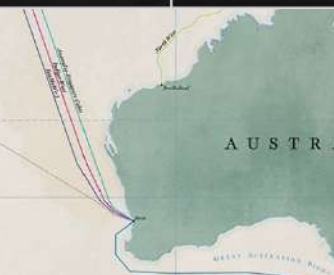
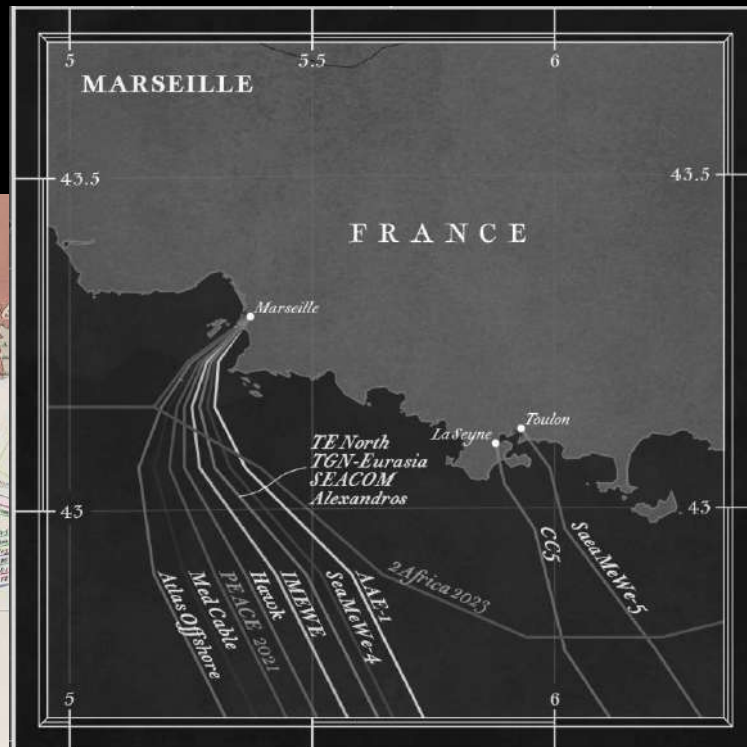
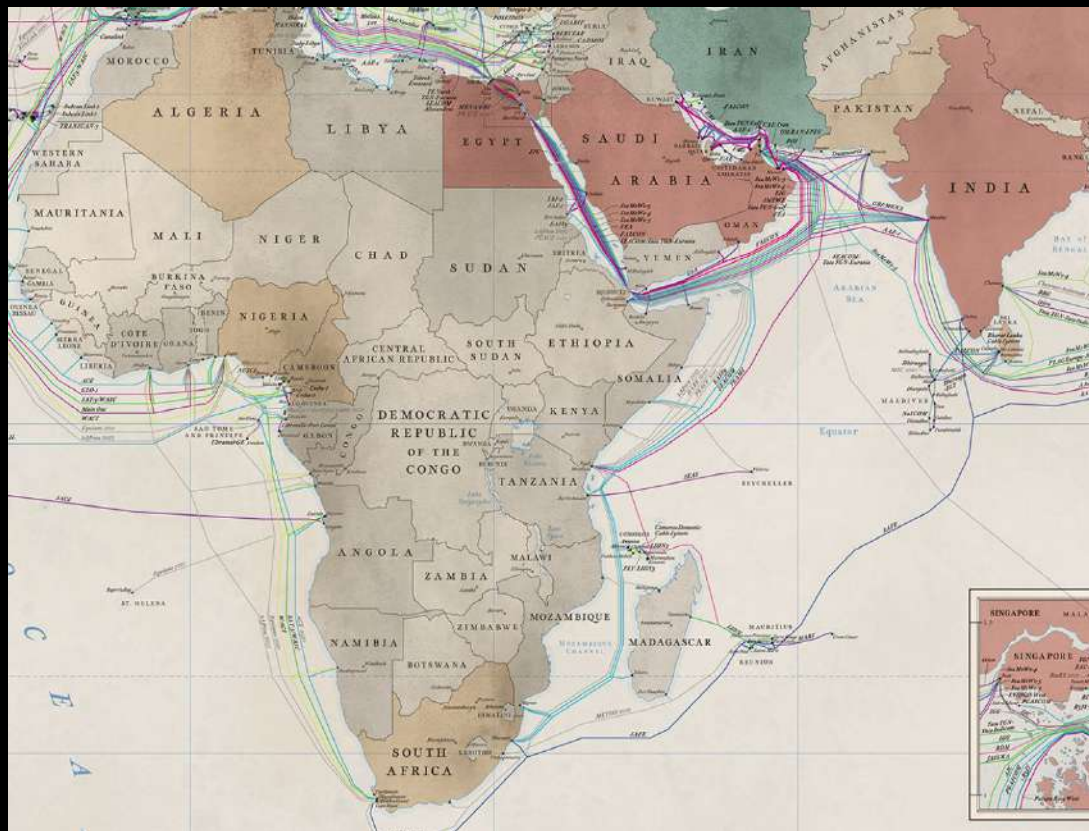


what is the internet, really?, andrew blum

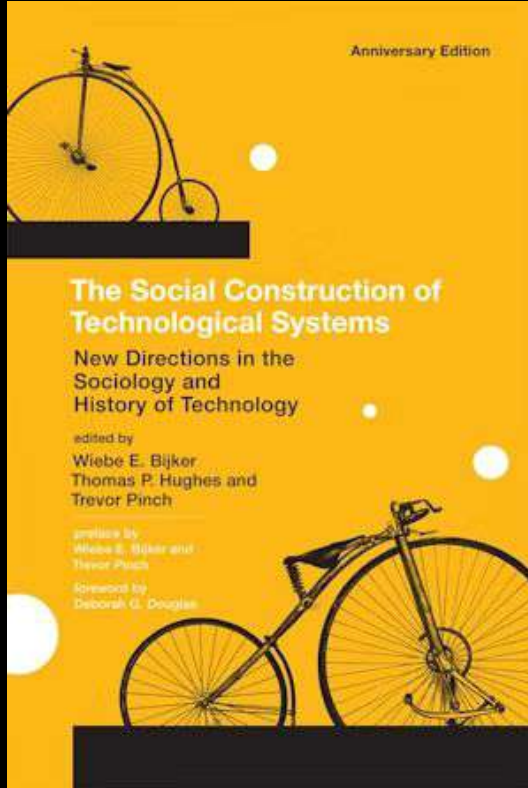
hudson 60 (western union building) →



une géopolitique sous-marine



de larges systèmes technologiques



thomas hughes, 1989 | 'the evolution of large technological systems',
in bijker, w.e., hughes, t.p. pinch, t.j. *the social construction of
technological systems. new directions in the sociology and history of
technology.* (pp. 51-82) massachusetts: mit press

“les systèmes technologiques contiennent des composants
désordonnés, complexes et destinés à résoudre des problèmes.
ils sont à la fois construits socialement et façonnent la société.”

les infrastructures sont

- de « large technological systems » - inextricablement
reliés à d'autres acteur·rices et réseaux technologiques,
sociaux, politiques et économiques.
- conçus pour ne pas être visibles
- des technologies perfectionnées jusqu'à devenir
routinisées ; on ne les remarque qu'en cas de défaillance.

quand est une infrastructure ?

1.1. When is an Infrastructure?

“What can be studied is always a relationship or an infinite regress of relationships. Never a ‘thing.’ ” —Gregory Bateson

Yrjö Engeström, in his “When Is a Tool?,” answers the implied title question in terms of a web of usability and action (1990). A tool is not just a thing with pre-given attributes frozen in time—but a thing becomes a tool in practice, for someone, when connected to some particular activity. The article is illustrated by a photo of a

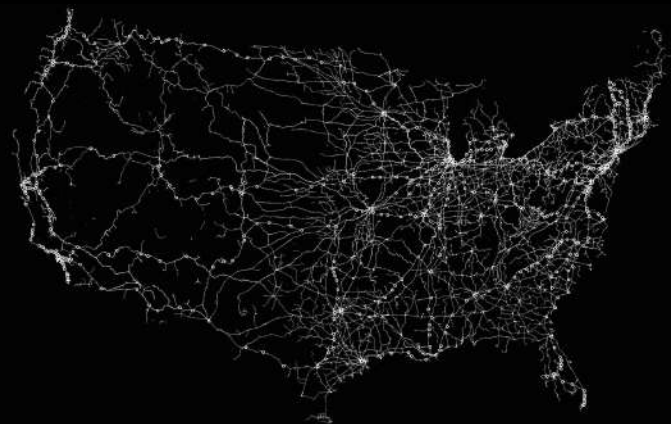
susan leigh star et karen ruhleder. “steps toward an ecology of infrastructure: design and access for large information spaces.” *information systems research*, vol. 7, no. 1, 1996, pp. 111–34.

l'infrastructure est un concept fondamentalement relationnel, elle devient infrastructure en relation avec des pratiques organisées. nous nous demandons donc quand, et non pas ce qu'est une infrastructure.

une infrastructure apparaît lorsque la tension entre le local et le global est résolue. autrement dit, une infrastructure apparaît lorsque les pratiques locales sont rendues possibles par une technologie à plus grande échelle, qui peut alors être utilisée de manière naturelle et immédiate. elle devient transparente à mesure que les variations locales sont intégrées dans les changements organisationnels

il ne s'agit pas d'un lieu physique ni d'un lieu permanent, mais d'une relation de travail

internet, l'infra- structure parfaite ?



réseau ferroviaire



réseau de fibre optique

à cet égard, l'internet mondial est l'infrastructure parfaite : il est omniprésent et invisible, partout et nulle part | grâce à lui, nous pouvons nous connecter à n'importe qui, n'importe où, depuis n'importe où, sans qu'il n'empiète sur notre réalité matérielle. | dans notre société postindustrielle, les technologies de l'information rendent les lieux et les espaces de plus en plus insignifiants | la distance n'existe plus, le monde est plat, l'unique est proche

ensmenger 2018 | the environmental history of computing
technology and culture vol. 59, 4:s7-s33.



FIG. 2 Bitcoin activity mapped against nuclear power plants, electrical power distribution grid. (Source: Image by author based on Google Maps.)

paris ville free |



2181 PARIS-SOUTERRAIN Les Egouts, Service de l'Assainissement; Collage de l'Agence de l'Eau Paris-Seine

iliad (free) acquiert en 2019 un abri anti-atomique de 8000 m2 situé à 25 mètres sous-terre (dc4)



SOMMAIRE

AVANT-PROPOS → 3

INTRODUCTION
PARIS VILLE FREE → 5
ENQUÊTE

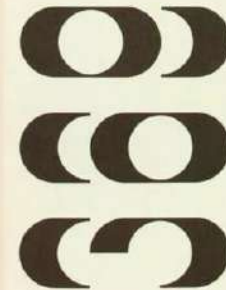
PARTIE 1
PARIS VILLE NUMÉRIQUE → 16
AUTOBIOGRAPHIE → 19
L'ODYSSÉE D'ILIAD → 23
FREE CENTER → 27
HOTLINE → 31
MÉGABITS PAR SECONDE → 35
FIBER TO MY HOME → 41
NŒUDS → 45
TOITS → 53
CLOUD → 59
RECYCLAGES → 67

PARTIE 2
REINVENTION → 70
BOSS → 73
PARIS OSE → 77
ÉCOLE → 81
DORTOIRS → 99
IMMERSION → 109

PARTIE 3
CÉLÉBRATION → 120
NEW FRANCE → 123
STATION → 125
COLOCATION → 141
MAISON → 147
COMMUNION → 151



Paris ville Free
— Soline Nivet





STANLEY TUMBLER

LE GAZ

ALICE
ALICE
ALICE

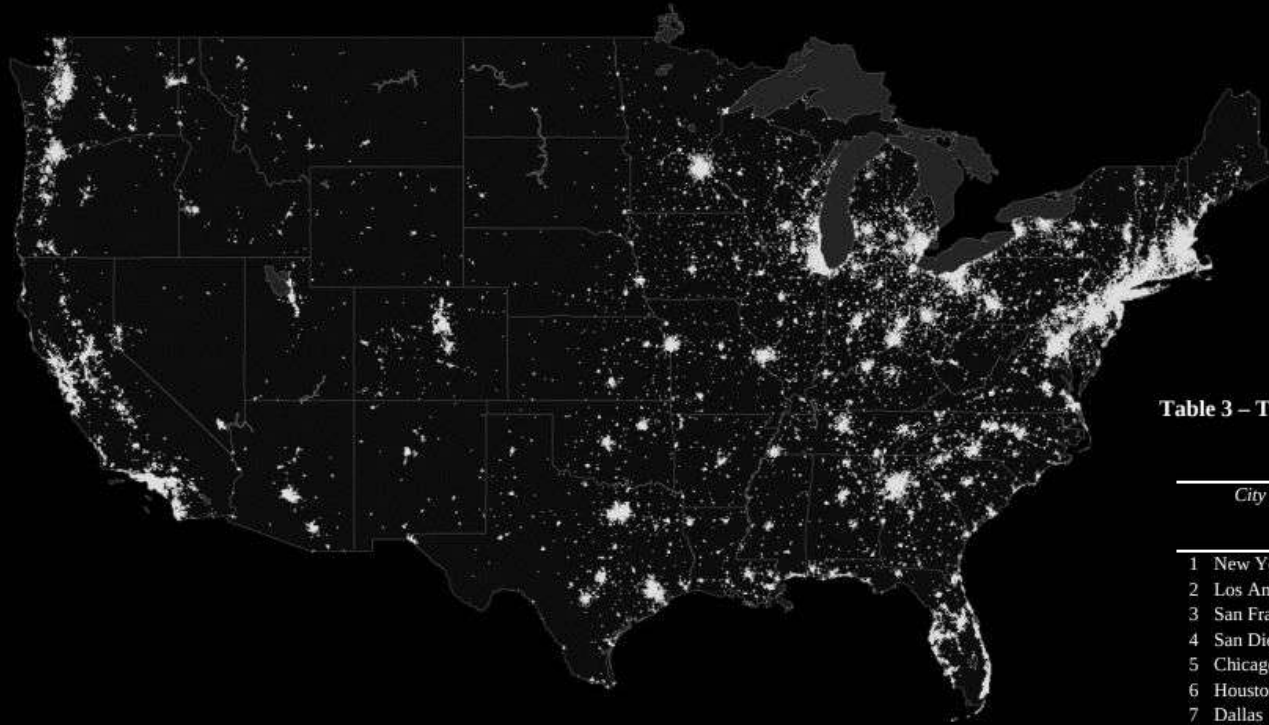
A vendre
CÉLESTINS
04 78 37 39 04



une infrastructure camouflée



1 5 | enjeux sociotechniques



matthew zook, 2000 |
 'the web of production: the
 economic geography of
 commercial internet content
 production in the united states'
*environment and planning a:
 economy and space*, 32(3),
 411-426.

Source: Domain Name data from Author Survey – June/July 1998

Table 3 – Top 15 Concentrations of Commercial Domain Names
 by Census Defined Cities, 1998

	City Name	State	Commercial Domains	Domains per 1000 Firms	Firm Specialization Ratio
1	New York	NY	56,003	310.11	1.42
2	Los Angeles	CA	38,086	328.39	1.51
3	San Francisco	CA	20,868	668.25	3.07
4	San Diego	CA	16,590	421.57	1.94
5	Chicago	IL	14,274	225.48	1.04
6	Houston	TX	13,822	192.34	0.88
7	Dallas	TX	12,074	215.72	0.99
8	Seattle	WA	10,754	446.46	2.05
9	San Jose	CA	9,883	359.45	1.65
10	Phoenix	AZ	8,588	226.27	1.04
11	Boston	MA	8,078	377.74	1.73
12	Portland	OR	6,606	257.81	1.18
13	Washington	DC	6,000	320.67	1.47
14	Austin	TX	6,000	293.63	1.35
15	Denver	CO	5,869	170.38	0.78

Source: Domain Name data from Author Survey – June/July 1998; Firm data
 is from 1995 Zip Code Business Patterns

une géographie de clusters
 qui renforce les inégalités

How Much Electricity is Needed to Mine 1 BITCOIN?

2024'S POST-HALVING

Legend:

- Least Expensive (Green)
- Most Expensive (Red)

Scale: \$0K, \$50K, \$100K, \$150K, \$200K

Costs by Country:

- Norway: \$94.68K
- Iceland: \$102.62K
- Denmark: \$258.21K
- Germany: \$269.47K
- Netherlands: \$212.53K
- Belgium: \$280.06K
- UK: \$271.46K
- Switzerland: \$236.37K
- Portugal: \$146.32K
- Spain: \$162.21K
- Italy: \$306.55K
- France: \$190.02K
- Sweden: \$160.23K
- Latvia: \$197.96K
- Finland: \$128.74K
- Estonia: \$189.96K
- Czech Republic: \$228.42K
- Iran: \$1.32K
- Poland: \$156.91K
- Ukraine: \$42.37K
- Kazakhstan: \$32.44K
- Romania: \$117.85K
- Uzbekistan: \$15.23K
- Kyrgyzstan: \$9.61K
- Russia: \$39.73K
- Syria: \$3.97K
- South Korea: \$66.73K
- Japan: \$143.01K
- Taiwan: \$60.91K
- China: \$50.98K
- Philippines: \$138.38K
- Hong Kong: \$125.13K
- Singapore: \$156.91K
- Indonesia: \$61.57K
- New Zealand: \$131.76K
- Australia: \$161.09K
- Cameroon: \$99.31K
- Thailand: \$79.45K
- Malaysia: \$84.75K
- Uganda: \$114.54K
- Kenya: \$163.54K
- India: \$50.98K
- South Africa: \$31.76K
- Botswana: \$9.93K
- Angola: \$8.61K
- Nigeria: \$9.27K
- Libya: \$5.29K
- Namibia: \$95.94K
- South Africa: \$122.49K
- Zimbabwe: \$181.41K
- Mozambique: \$84.09K
- Ethiopia: \$1.99K
- Burkina Faso: \$3.97K
- Argentina: \$12.58K
- Uruguay: \$160.23K
- Chile: \$119.18K
- Peru: \$116.53K
- Ecuador: \$63.56K
- Jamaica: \$181.41K
- Cayman Islands: \$268.81K
- Honduras: \$150.29K
- Costa Rica: \$115.87K
- Panama: \$101.96K
- El Salvador: \$153.60K
- Mexico: \$64.88K
- Bhutan: \$280.72K
- USA: \$107.26K
- Canada: \$84.09K
- Ireland: \$321.11K

carbonate de lithium à *salar de uyuni* (bolivie, près de potosi)

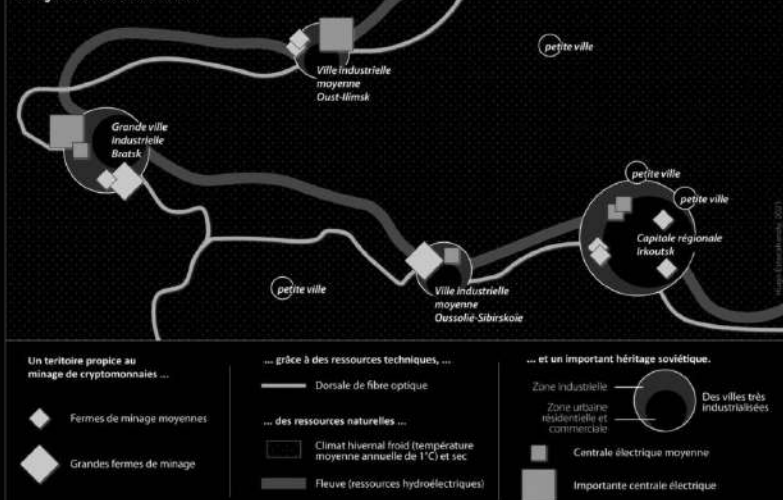
miner du bitcoin sur l'héritage soviétique

hugo estecahandy
dans *reset* | 2025

Les enjeux du minage de cryptomonnaies en Asie centrale, sur un réseau électrique commun diversifié mais désuet

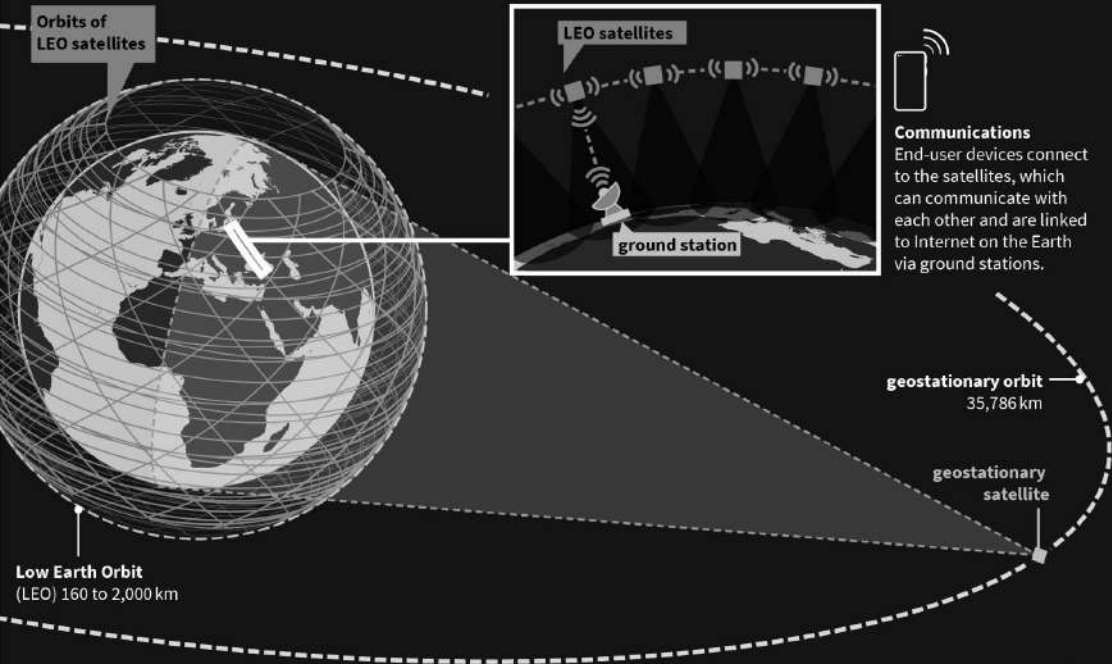


Modélisation des logiques de localisation des fermes de minage dans l'oblast d'Irkoutsk



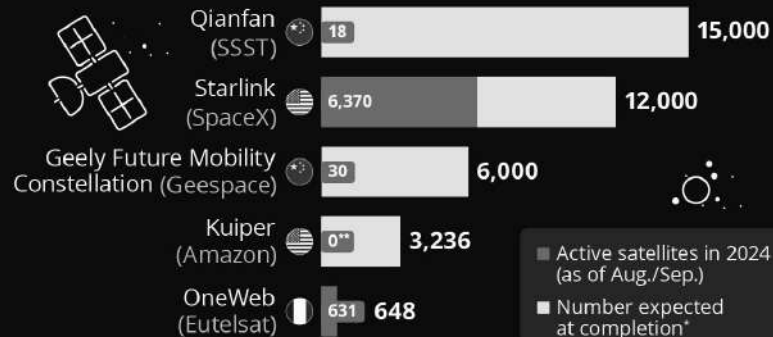
le nouveau far west de l'orbite terrestre basse

Satellites in Low Earth Orbit (LEO) Schematic representation



Satellite Constellation Projects Ready for Takeoff

Major satellite constellation projects currently being deployed worldwide, by number of (active) satellites



* Extensions announced: up to 40,000 (Starlink), more than 7,000 (Kuiper)

** Two prototypes already launched (project to start 2025)

Sources: Media reports, CelesTrak

un dangereux monopole oligarchique infrastructurel



INFRASTRUCTURE

La panne du service satellite Starlink affecte l'armée ukrainienne

soldats ukrainiens sur la ligne de front. © Anadolu via Getty Images

Rédaction Data News
15-09-2025, 14:40 • Mise à jour le: 15-09-2025, 17:51 • Source: ANP • < 1 min. de lecture

THE KYIV INDEPENDENT
Wednesday, September 17, 2025

Europe Video Culture Investigations War Crimes Dare to Ukraine Explaining Ukraine

NEWS FEED

Musk ordered Starlink shutdown during Ukraine's 2022 Kherson counteroffensive, Reuters reports

July 25, 2023 5:20 pm • 2 min read

by Tim Zatorsky



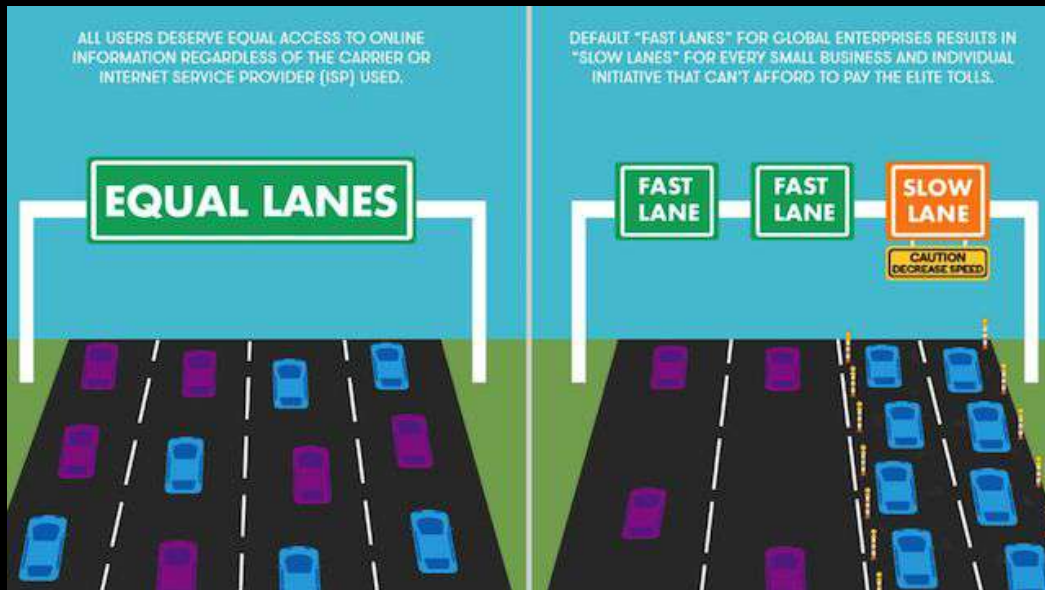
A portrait of Elon Musk taken in Krakow, Poland, on Jan. 22, 2023, alongside Russia's national flag, is shown for illustration. (Beata Zawrzejko/Photo via Getty Images)

régulation d'internet et neutralité du net

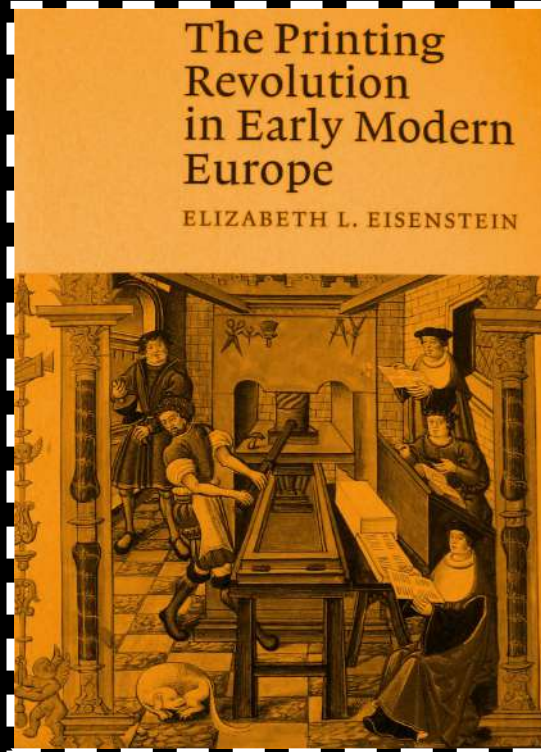
- gouvernance internationale :
- IETF (Internet Engineering Task Force)
 - ICANN (Internet Corporation for Assigned Names and Numbers)
 - W3C (World Wide Web Consortium)

THIS COULD BE THE INTERNET WITHOUT NET NEUTRALITY.

PACK 1 \$5 MONTH	PACK 2 \$10 MONTH	PACK 3 \$15 MONTH
GET ACCESS TO: f t w	GET ACCESS TO: f t w i	GET ACCESS TO: f t w i y v n
PACK 4 \$20 MONTH	PACK 5 \$30 MONTH	UNLIMITED \$100 MONTH
GET ACCESS TO: f t w i y v n	GET ACCESS TO: f t w i y v n +	GET ACCESS TO EVERYTHING



conclusion |



un déterminisme technologique ?

- l'infrastructure d'internet a une dimension profondément politique ; nier sa matérialité est écologiquement extrêmement néfaste
- nous vivons une rupture technologique, mais aussi en termes de propriété, de gouvernance et des valeurs qui régissent le développement et la régulation de cette dernière